

Question 15: Analyze the stability and efficiency of Merge Sort by executing it on datasets containing duplicate values. Observe how the relative order of equal elements is maintained. Record execution time and compare with Quick Sort. Interpret the results and justify why Merge Sort is considered stable.

```
EXECUTION.py - C:/Users/jasmi/OneDrive/Desktop/DAA LAB EXP/EXECUTION.py (3.14.4)
File Edit Format Run Options Window Help

import time

# ----- Merge Sort -----
def merge_sort(arr):
    if len(arr) <= 1:
        return arr

    mid = len(arr) // 2
    left = merge_sort(arr[:mid])
    right = merge_sort(arr[mid:])

    result = []
    i = 0
    j = 0

    while i < len(left) and j < len(right):
        # <= keeps equal elements in their original relative order
        if left[i][0] <= right[j][0]:
            result.append(left[i])
            i += 1
        else:
            result.append(right[j])
            j += 1

    result.extend(left[i:])
    result.extend(right[j:])

    return result

# ----- Quick Sort -----
def quick_sort(arr):
    if len(arr) <= 1:
        return arr

    pivot = arr[len(arr) // 2][0]

    left = [x for x in arr if x[0] < pivot]
    middle = [x for x in arr if x[0] == pivot]
    right = [x for x in arr if x[0] > pivot]

    return quick_sort(left) + middle + quick_sort(right)
```

```

# Dataset containing duplicate values
data = [
    (5, 'A'),
    (3, 'B'),
    (5, 'C'),
    (2, 'D'),
    (3, 'E'),
    (5, 'F'),
    (2, 'G'),
    (4, 'H')
]

print("Original Data:")
print(data)

# Merge Sort
start = time.perf_counter()
merge_result = merge_sort(data.copy())
merge_time = time.perf_counter() - start

print("\nMerge Sort Result:")
print(merge_result)
print("Merge Sort Time:", merge_time, "seconds")

# Quick Sort
start = time.perf_counter()
quick_result = quick_sort(data.copy())
quick_time = time.perf_counter() - start

print("\nQuick Sort Result:")
print(quick_result)
print("Quick Sort Time:", quick_time, "seconds")

```

File Edit Shell Debug Options Window Help

Python 3.14.4 (tags/v3.14.4:23116f9, Apr 7 2026, 14:10:54) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>

===== RESTART: C:/Users/jasmi/OneDrive/Desktop/DAA LAB EXP/EXECUTION.py =====

Original Data:

[(5, 'A'), (3, 'B'), (5, 'C'), (2, 'D'), (3, 'E'), (5, 'F'), (2, 'G'), (4, 'H')]

Merge Sort Result:

[(2, 'D'), (2, 'G'), (3, 'B'), (3, 'E'), (4, 'H'), (5, 'A'), (5, 'C'), (5, 'F')]

Merge Sort Time: 1.7699989257380366e-05 seconds

Quick Sort Result:

[(2, 'D'), (2, 'G'), (3, 'B'), (3, 'E'), (4, 'H'), (5, 'A'), (5, 'C'), (5, 'F')]

Quick Sort Time: 1.3199984095990658e-05 seconds

>>>

|